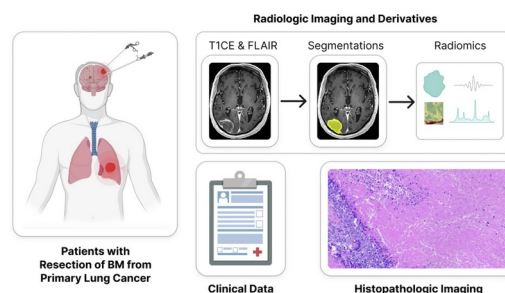


Quantifying Microscopic Tumor Extension Beyond the Radiographic Margin in Brain Metastases: A Multimodal MRI–Histopathology Correlation Study



Brain metastases are conventionally assumed to have a sharp, well-circumscribed boundary between tumor and surrounding brain tissue, unlike infiltrative primary brain tumors such as glioma. This assumption underlies clinical practice, including the narrow margins (0–2 mm) used in stereotactic radiosurgery planning, yet it has rarely been rigorously quantified using matched imaging and tissue data. This thesis uses a public dataset of 111 matched cases (T1-contrast-enhanced and FLAIR MRI, tumor segmentations, radiomic features, and whole-slide H&E histopathology) from brain metastases of primary lung cancer to measure the distance and pattern of microscopic tumor cell infiltration beyond the radiographically-defined tumor margin, and to test whether peritumoral radiomic features (derived from FLAIR hyperintensity) can predict infiltrative versus sharply demarcated growth patterns on histology. As a secondary aim, the thesis explores whether computational features extracted from the H&E slides can predict clinically-relevant biomarkers already available in the dataset (PD-L1 status, histologic subtype), providing a low-cost, image-based complement to standard immunohistochemistry. The results will contribute quantitative evidence relevant to defining safer, more personalized radiotherapy margins in brain metastasis management.

Focus:

Image processing and machine learning

Data:

will be provided as NIFTI and TIFF stack.

Starting code: not available, but look for inspiration any repository about the BRATS challenge.

Requested background: Python and image processing

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